Nudge a local-AI assistance dog for deaf, blind and deafblind owners

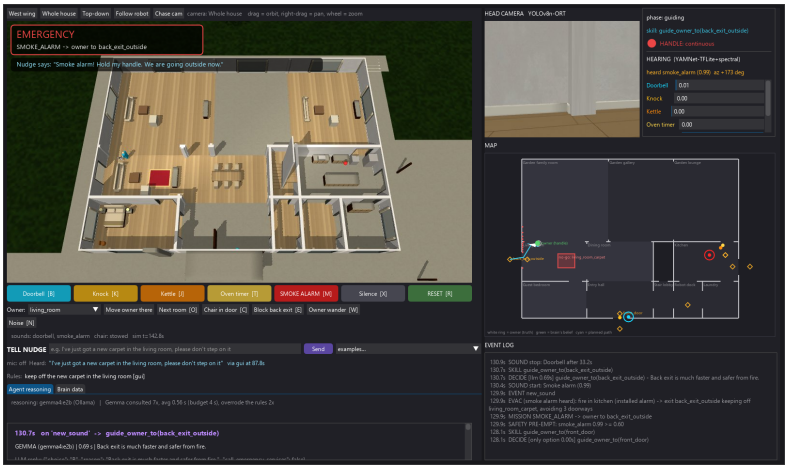
Agentverse 2026 · Robotics track · Unitree Go2 in MuJoCo · 100 % local AI

Rules for safety. Local Gemma for the unknown.

Why it matters in the UK. Millions of people live with sight or hearing loss, yet good free care and social support are hard to get, so many feel unsafe and isolated in their own homes. A trained assistance dog can change that, but there are far too few: a guide dog takes about two years to train and the wait to be matched is up to 18 months, with some owners going over two years without one. Sense found that **1 in 5 deafblind people** received no social care at all, and those who did needed on average **30 % more hours** than they got. In between, the doorbell goes unanswered and a smoke alarm can go unheard.

2 million+	**18 million**	**450,000+**	**~900**
people in the UK living with sight loss (RNIB)	deaf, with hearing loss or tinnitus: 1 in 3 people (RNID)	deafblind people, rising to 610,000+ by 2035 (Sense)	people waiting for a guide dog; waits up to 18 months (ITV, Jan 2026)

Nudge is an always-available, low-cost assistance robot: a Unitree Go2 that listens to the house, works out what a sound is and where, finds its owner with its camera, **physically nudges them**, tells them through every channel they can perceive (handle vibration, light bar, screen, speech) and **leads them by a rigid guide handle** to the door, to the kettle, or out of the house by the safest exit in a fire. Every decision comes from live sensor data, never a scripted path.



Live run, unscripted change: mid-way through guiding the owner to the front door, a smoke alarm goes off. Plain-code safety pre-empts the mission (0.6 s); local Gemma picks the back exit ("much faster and safer from fire", 0.69 s) and the route still avoids the carpet the owner asked it not to step on. Left: MuJoCo 3D view. Right: head camera + YOLO, hearing, map, event log.

How it works: sense → reflex → think → act

- **Two processes over ZeroMQ**, so 500 Hz MuJoCo physics never waits for AI inference.
- **Senses (simulated like real hardware):** head RGB-D camera, 36-ray 2D LiDAR, stereo ears (distance + wall attenuation, level/time differences between ears, 5 background-noise profiles).
- **Perception:** YAMNet + spectral rules (what the sound is), stereo bearing over several robot poses (where it is), YOLOv8n (the owner), a Bayes filter over which room the owner is in.
- **Reflex layer, plain code:** smoke alarm pre-empts everything; exits checked against the fire position; LiDAR obstacle stops; 0.4 m/s cap near the owner; 4 s watchdog on every LLM call. The LLM can never override it.
- **Thinking layer, local Gemma 4 E2B:** on each event it picks one option from a menu of *legal* actions built from live perception, with a one-line reason on screen. JSON-schema constrained, so it cannot produce an invalid action; rules take over if it is slow or offline.
- **Talk to it:** "Nudge, I've got a new carpet, please don't step on it" (local Whisper) becomes a rule the planner enforces. Teach jobs in one sentence ("every night at 10, take me to bed"); requests it can't do ("call my daughter") are refused, never promised.
- **Owner profiles:** deaf / blind / deafblind choose which outputs fire; one haptic code per urgency tier, so a deafblind owner learns three patterns, not a dozen.

Adapts when the scene changes

- **Owner moves mid-mission:** the room belief updates from what the camera did and didn't see; it re-searches by probability and never rescans a room it already cleared (owner found ~6.5 s sooner).
- **Owner speaks:** it hears *where* the voice came from and searches that room first.
- **Is the sound real?** A doorbell on the TV is told apart from the front door by direction + loudness: "That was the TV, not the door."
- **Fire-aware evacuation:** picks the exit by walk length + distance from the fire; if LiDAR sees the chosen exit blocked, it re-chooses mid-evacuation. Asks "Shall I call 999?" while still leading the owner out.
- **Obstacles + getting stuck:** chair in a doorway → LiDAR marks it blocked and re-plans; no progress → back off, remember the obstacle, re-route.
- **Owner doesn't respond:** repeated unanswered nudges → alerts the emergency contact and stays.
- **Overlapping sounds:** kettle during a doorbell mission is deferred, then handled; a smoke alarm interrupts anything.
- **A home it has never seen:** a second home (apartment) runs unchanged, and the robot can explore a house on its own and build its own map, objects and exits.

Also built, off by default: fallen-owner detection (depth + Gemma camera check → "Are you OK?" → 999), unknown sounds reasoned by Gemma (glass, dog, baby), LiDAR particle-filter localisation.

Measured (all local, headless sim)	Result
Gemma 4 E2B decisions (Ollama, RTX 5060 laptop)	**0.19 s** median, 0.26 s p95; **100 %** inside the watchdog; 93 % right on real logged decisions
Smoke alarm pre-emption	interrupts any mission in **0.6 s**; owner led outside in every evacuation scenario
Hearing, 1,000 random trials (5 noise profiles, random room + distance)	registry sounds alerted **96 %**, median 0.5 s; **smoke alarm never missed**; background noise alone **0 %** false alerts
Owner detection, YOLOv8n on the head camera	**97–98 %** of 412 views (6 rooms × 4 headings × 3 distances), median position error 6 cm
TV-vs-door check	0 of 243 real door sounds mistaken for the TV; a real doorbell is not delayed
Robot's own map (frontier exploration)	**99.8 %** of the floor found; walls 97.4 % precise; exits found within 0.01–0.21 m
Teaching by sentence	**19 / 19** test sentences parsed to the right task or refused
Scenario suite (ground-truth judge)	default suite **14 / 14** PASS; robot-map mode 22 / 24 + **9 / 9** soak; apartment doorbell, smoke, kettle PASS

Every model runs on the laptop: Gemma 4 E2B via Ollama (GPU) · YOLOv8n ONNX (CPU, ~60 ms/frame) · YAMNet TFLite (CPU) · Whisper base.en (CPU, ~0.5 s/sentence) · Piper TTS (CPU). SHA-256-pinned, one fetch command. **Honest limits:** pose comes from the simulator by default (particle filter built, off); walking is kinematic, not a learned gait; some furniture is proxy boxes; real household recordings are weaker than the demo clips (smoke 96 % vs 100 %), so the demo uses synthesized sounds.

Sources: RNIB sight-loss stats; RNID hearing-loss prevalence (2026); Sense, Deafblindness; ITV News 27 Jan 2026 (guide dogs); BBC News 18 Jun 2023; Sense to Parliament (EXP0029).
Team Nudge: Eden, Finn, Ricktho, Sean, Teo · Scene: nerv-world house (MIT), Go2 from MuJoCo Menagerie (BSD-3) · Numbers measured headless in simulation on 2026-11-08